

MATH 351 – Probability

Spring 2025

Instructor: James Parson

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Sending e-mail is a good way to contact me. I'm happy to answer questions. When in doubt, write me! I generally answer e-mail within 24 hours.

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Office hours: Mondays 10:30 am – 12 noon, Tuesdays 10:00 am – 11:30 am and 3:00 pm – 5:00 pm, Wednesdays 12:30 pm – 1:30 pm.

These are the times when I guarantee that I will be in my office and available to discuss anything related to our class. Please stop by to see me! I love to talk about probability. If you cannot make it to my office during these times, we can make an appointment to talk at another time. We can set that up with a short chat after class, or we can talk by e-mail.

Class meeting time and location: Tuesdays and Thursdays 8:15 am – 9:40 am, Hodson Science Center room 316

Course description and purpose

Catalog description

A calculus-based course on the theory and application of modern probability. Topics include events and probabilities, random variables and distributions, expectation, conditional probability and independence, the Central Limit Theorem.

Prerequisites

MATH 112 or equivalent and MATH 202. (Courses equivalent to MATH 112 for prerequisite purposes include ECMG 212, MATH 213, PSY 211, and SOC 261.)

Course purpose

The Mathematical Association of America's Committee on the Undergraduate Program in Mathematics writes, "Probability courses teach students, above all, how to think systematically about uncertainty and randomness." Here is the sort of thing that the authors of that sentence have in mind: there are some events in the world that seem totally unpredictable; for example, if someone flips a coin, there is no telling whether it will land on "heads" or "tails." On the other hand, if that person flips a coin a hundred times, probability theory tells us not only that we should expect about 50 "heads" but, more precisely, that we should expect 50 "heads" give or take 5 heads—and that seeing fewer than 40 "heads" should be rather surprising. The "think[ing] systematically" is learning how to reason like this about uncertain events; although we cannot remove the uncertainty, we can quantify it and calculate with it.

Probability theory makes it possible to quantify and manage risk. Insurance companies help their policyholders diffuse individual risks by sharing them in larger pools. Investors craft mixed market positions that balance high-yield risks with low-yield sure bets. Casinos and state lottery commissions finely tune their games to tantalize their customers while virtually ensuring their long-run profitability. Probability theory also undergirds the science of statistics, used universally to make sense of the ever-more abundant data in our world.

In this course, we will study the foundational ideas of probability theory, learning its characteristic style of reasoning and practicing solving a wide variety of practical problems. This course can be used to fulfill a course requirement within the mathematics major and can contribute to a mathematics minor. It is a required course for students pursuing any of the actuarial science minors.

Goals and objectives

Program level goals and objectives

Graduates earning bachelor's degrees in mathematics from Hood College

1. Demonstrate skill in problem solving through identifying appropriate mathematical techniques and implementing techniques effectively to reach a solution.

Course specific goals

I have the following hopes and expectations for you for this semester: after completing this course, you will...

- Be able to describe the basic formalism of probability theory, including sample spaces, events, the axioms of probability, and random variables and use that formalism to reason systematically about uncertain events.
- Recognize some frequently used discrete and continuous probability distributions and use those distributions to model real-world situations.
- Solve problems using conditional probability and conditioning (major goal: problem solving).
- Solve problems in discrete probability with combinatorial techniques.
- Solve problems in continuous probability with calculus techniques.
- Understand better the role of modeling in applied mathematics and develop comfort with the role of risk when modeling for real-life applications.

Format and procedures

Our class meetings will involve a mixture of activities, which I hope will keep you engaged with the course material. I will give some short lectures on course topics. I hope that we will spend a lot of time, though, working in small groups on various activities related to whatever topic we are studying. When you are working on these activities, we will switch back and forth between group work and class discussion so that you can share what you have figured out and ask questions about tricky points. These activities will generally follow the textbook closely. It will often take us several days to work through a section of the text.

I will not grade the activities that we do together in class, but working through them carefully will benefit you in several ways. First, working on those activities is how you will learn the subject matter of the course. You will apply your knowledge right away to solve the homework problems that I assign associated with each topic. Most weeks, we will also have a short group quiz. These quizzes will include a few questions related to the activities from

previous days' classes. You will work with a small group (maybe just with one partner) on the quiz questions, and you will be allowed to refer to your textbook, your in-class work, and your homework while answering these questions. Finally, the questions in all three of our exams (two midterm exams and a final exam) will be modeled on the questions from our in-class activities: your familiarity with those in-class learning activities will help you succeed on these exams.

Because of the importance of the in-class activities to your success in meeting the course objectives, it is vital for you to attend all the class meetings. You should expect to participate actively in class every day, and since most of the learning activities involve group work, you should be ready to collaborate, being respectful of your group members and doing your share of the work.

Course expectations and requirements

Course texts and other resources

Our primary course text is

Durrett, Rick. 2009. *Elementary probability for applications*. Cambridge: Cambridge University Press.

The author is an expert on probability theory. He has written many books on the subject, of which our textbook is the most elementary. (The others include a graduate-level textbook on probability theory and several books on “stochastic processes.”)

We will use the Durrett's many excellent examples and problems in class work and for homework.

I'll warn you that Durrett's book is old—he probably wrote much of it in the 1990s—and it has what we might generously describe as quirks. Here are things you'll notice this semester:

- Exercises may refer to antique technology such as telephone books and magnetic tapes.
- Durrett can be casually sexist.
- Durrett must have gotten a TI-83 calculator just as he was writing the book. He is very excited to tell us how to use one.

A secondary purpose of this course is to help prepare students interested in actuarial science for the probability actuarial exam. You can find information about this exam at

<https://www.soa.org/education/exam-req/edu-exam-p-detail.aspx>

Blackboard

Blackboard is the learning-management system that Hood uses for all courses. I will use Blackboard to make course announcements, to share homework assignments, to report your grades, and more.

In addition to bringing paper copies of activities to class, I will post electronic copies to Blackboard. You are welcome to bring your own laptop or other device to class for Blackboard access if you prefer to use the electronic versions rather than paper versions.

Calculator

You do not need to buy a special calculator for this class. Any calculator that you have will be fine. I also have calculators that students can borrow during class time.

Although you may use your phone as a calculator during class activities, you may not use it for tests. If you want to use a calculator during a test and do not have your own calculator, you can use one of the mathematics department's calculators.

Homework

Learning any kind of mathematics is like learning a language in that it takes regular practice. To help you practice, I will assign homework after most classes. Keeping engaged with the course topics by doing homework is an important part of the course. You may think of homework as a bridge between the in-class activities and the three exams. The best way to be prepared for success on the exams is to work hard on the regular homework assignments.

Homework is also an important way for us to communicate. When I read what you turn in, I learn what points are clear to you and what points are still challenging. Learning these things helps me decide what extra practice to give you. Homework also lets me give you feedback so that you also know which skills to focus on improving.

You will find all homework assignments in the "Assignments" folder of our Blackboard page, and you will turn in your completed homework electronically in the same part of Blackboard. You should check Blackboard regularly for these assignments, even if I do not specifically announce that I will post an assignment. Homework is usually due by the start of the next class after I assign it.

You may collaborate on homework assignments with other students, but you must write up the work that you turn in on your own. If I find homework submissions that are practically identical, I will not accept them for credit. The reason for this policy is that completing the homework is the way you start practicing the skills that show up everywhere in the course. You need to try these things yourself, not only to learn to do them but also to recognize whether you know how to do them. It is easy to trick yourself when you read something that another person has written. You may think that you could do it yourself just because it makes sense on the page. Writing your own homework helps you judge how well you understand the assignment's skills.

I value your questions about homework assignments. While you are working on the homework, you may send questions by e-mail (which I will try to answer promptly), or you may ask questions during my office hours. I will also take questions that you have about homework problems at the start of class.

We will start most class days with a discussion of the daily homework assignment. You will first talk over the assignment in pairs or with a small group of students. After this small-group discussion, we will talk about the assignment as a whole class. I may ask groups to present solutions to some problems, including any problems where students have questions. You will use what you learn in our homework discussion to revise your work so that you can improve and resubmit it (as explained below).

How to turn homework in

You will submit your homework to me using links on the "Assignments" tab of our Blackboard page. You must submit your work before class so that I can see what you understood prior to our group discussion.

After we discuss the homework, you may revise your work and resubmit it on Blackboard. Any resubmission is due by 11:59 pm on the day that the homework is due. As long as your first submission is a solid attempt at the assignment, I will assign your homework grade based on the revised version.

Late homework

I will accept late homework, but only until 11:59 pm on the day that the homework is due (when homework revisions are also due). You should expect a late penalty of 20% of the available points. If there are external factors keeping you from turning in a homework assignment on time (e.g. illness or family emergency), please tell me so that I can either exempt you from the assignment or waive the late penalty.

When computing your course grade, I will drop your two lowest homework grades.

Group quizzes

Most weeks, we will have a group quiz about the topics we are exploring in class. You will complete these quizzes in pairs or in small groups. During the group quizzes, you will be allowed to use the following:

- the textbook,
- resources that I have posted on Blackboard,
- and your own notes.

We will use these group quizzes to focus on tricky points that come up in class activities and homework. The quizzes will provide practice with the sorts of questions that I will ask in our midterm exams and final exam.

Please tell me if there are external factors (e.g. illness, family emergency, Hood-sponsored activity such as field trip or athletic event) that will prevent you from being in class for a quiz. Depending on the circumstances, I may offer you a make-up quiz, or I may exempt you from the quiz.

When computing your course grade, I will drop your lowest quiz grade.

I have scheduled quizzes for August 25th, September 2nd, September 9th, September 16th, September 23rd, October 2nd, October 9th, October 21st, October 28th, November 6th, November 13th, and November 20th.

Midterms and final exam

We will have two midterm exams and a final exam. I will allow you to use a sheet of notes for these exams. During the tests, you may use your own calculator, but you may not use your phone as a calculator. The mathematics department has calculators to loan to anyone who needs them for taking tests.

Before each of the two midterm exams, I will tell you which topics will be included in the test. The final exam will be comprehensive, including topics from the entire course.

I have scheduled the midterm exams as follows:

Midterm 1 will be on Thursday, September 25th.

Midterm 2 will be on Thursday, October 30th.

Our final exam will be on Thursday, December 4th 8:30 am – 11:00 am in our usual classroom.

Outside events

I expect you to attend at least one mathematics seminar or other event this semester and to type a short discussion of the event. I will share (and update regularly) a list of outside events available this semester.

Grades

I will determine your final grade using the following weights:

Two midterm exams	35%
Homework	30%
Quizzes	10%
Comprehensive Final Exam	25%

Outline of course topics

The following table includes an outline of the topics that we will discuss during the semester.

Week	Topics	Text	Exams
1	Course introduction and syllabus; counting techniques	Ch.2	
2	More on counting techniques; the formalism of probability	Ch. 2, Ch. 1	
3	Geometric distribution; expected value; moments/variance/standard deviation	Ch.1, Ch.2	
4	Binomial and multinomial distributions; Poisson approximation to binomial.	Ch. 2	
5	Urn problems; hypergeometric distribution; inclusion/exclusion		
6	Midterm Exam 1; Conditional probability	Ch. 3	Exam 1
7	Two-stage experiments and Law of Total Probability; Bayes' Rule.		
8	Joint distributions: marginal distributions, conditional distributions, independence of random variables (all discrete); continuous probability distributions and density functions. Fundamental examples: uniform, exponential, power.	Ch. 4	
9	Expected values; distribution functions; functions of random variables (transformations)		
10	Order statistics; independence of continuous random variables		
11	Midterm Exam 2; catch-up as needed		Exam 2
12	Expected value of sums; Bernoulli random variables; method of indicators; Applications to binomial, hypergeometric distributions	Ch. 6	
13	Gamma-distributed variables; Variance of sums; Law of Large Numbers		
14	Laws of Large Numbers; Normal distribution; Central Limit Theorem		
15	One class meeting (11/25) Applications of Central Limit Theorem; continuity correction		
16	One class meeting (12/2) More examples with Central Limit Theorem		
Finals week	Final exam on Thursday 12/4, 8:30 am – 11 am.		Final exam

Academic Honor Code

As a place of honor and respect, all members of Hood College assume the obligation to maintain the principles of honesty, responsibility, and intellectual integrity in all activities related to their Hood College experience. Students

are expected to adhere to the highest standards of academic honesty and integrity in all coursework and related matters. It is the responsibility of each student to support these values through maturity of thought, expression, and action. Members of the faculty and staff are available to assist students in this process. The Academic Judicial Council, which is composed of undergraduate students and two elected faculty members, is chaired by Dr. Jolene Sanders. If you have questions about the Academic Judicial Council or the Hood College Honor Code, please contact Dr. Sanders at sanders@hood.edu.

Counseling Services Information

If you are feeling stressed, worried, or down during the semester, or if you notice signs of emotional distress in someone else, you may visit or contact Amanda Dymek, director of wellness. Hood also has many campus resources that are available to support you including the following:

- Hood Counseling Services – open M – F, 8:30 am – 5:00 pm, Apple Resource building, 1st floor, Room A. Make your appointment online at hood.edu/counseling, or drop in for a walk-in session on Mondays, Wednesdays, and Fridays between the hours of 11:00 am – 12:00 pm. If you have questions about Counseling Services, email counselingservices@hood.edu
- Hood also has several connections to other mental health resources including the NeighborHood Counseling Training Center and Thriving Campus.

If you need to talk to someone right away, text or call 9-8-8 for a free, confidential conversation with a trained counselor.

Accessibility

This course is intended to be accessible for all students, including those with mental, physical, or cognitive disabilities, illnesses, injuries, and impairments. If you are not able to access fully the space, content, and experiences of this course, you are welcome (not required) to contact me by e-mail, phone, or during office hours to discuss your specific needs. You may also contact the Office of Accessibility Services (301-696-3569 or accessibilityservices@hood.edu). Accessibility Services can help you document your needs and create an accommodation plan. By making a plan through Accessibility Services, you can ensure appropriate accommodations without disclosing your condition or diagnosis to course instructors.

Title IX

Hood College faculty are committed to helping create a safe and open learning environment for all students. If you (or someone you know) have experienced any form of sexual misconduct, including sexual assault, dating or domestic violence, or stalking, know that help and support are available. The College strongly encourages all members of the community to take action, seek support, and report incidents of sexual misconduct to the Title IX Coordinator. Please be aware that under Title IX of the Education Amendments of 1972, I am required to disclose information about such misconduct to the Title IX office. If you wish to speak to a confidential employee who does not have this reporting responsibility, you can contact the Dean of the Chapel (chapel@hood.edu), counseling services (counselingservices@hood.edu), or health services (healthservices@hood.edu). For more information about reporting options and resources at Hood College and the community, please contact the Title IX Coordinator.

Inclusivity and campus values

Hood College is proud of its diverse community, and we are committed to cultivating and strengthening an inclusive, tolerant, multi-cultural, and intellectually open community with equal opportunity for all. By encouraging and celebrating our differences, we create an environment that promotes freedom of thought and academic excellence. It is our goal to have a respectful and nurturing academic community that affirms the inherent worth and dignity of every individual, and celebrates the diverse backgrounds of all students, faculty, and staff. We will strive to value each person for their uniqueness and difference and to encourage all community members to reach their fullest potential.

Student Success Services

Your success in my class is a priority. For this reason, I will be using Blazer Navigate as an early identification and intervention tool. If I notice that you are struggling with issues such as attendance, class participation, or assignment/test performance, I may send an academic alert through Blazer Navigate to connect you with appropriate campus resources. These referrals are designed to help you succeed; they are not a reprimand or punishment. Please respond to any communications you may receive from me, your academic advisor, the Student Success Center, your athletic coaches, the dean of student's office, or other campus offices regarding your academic progress in this course.